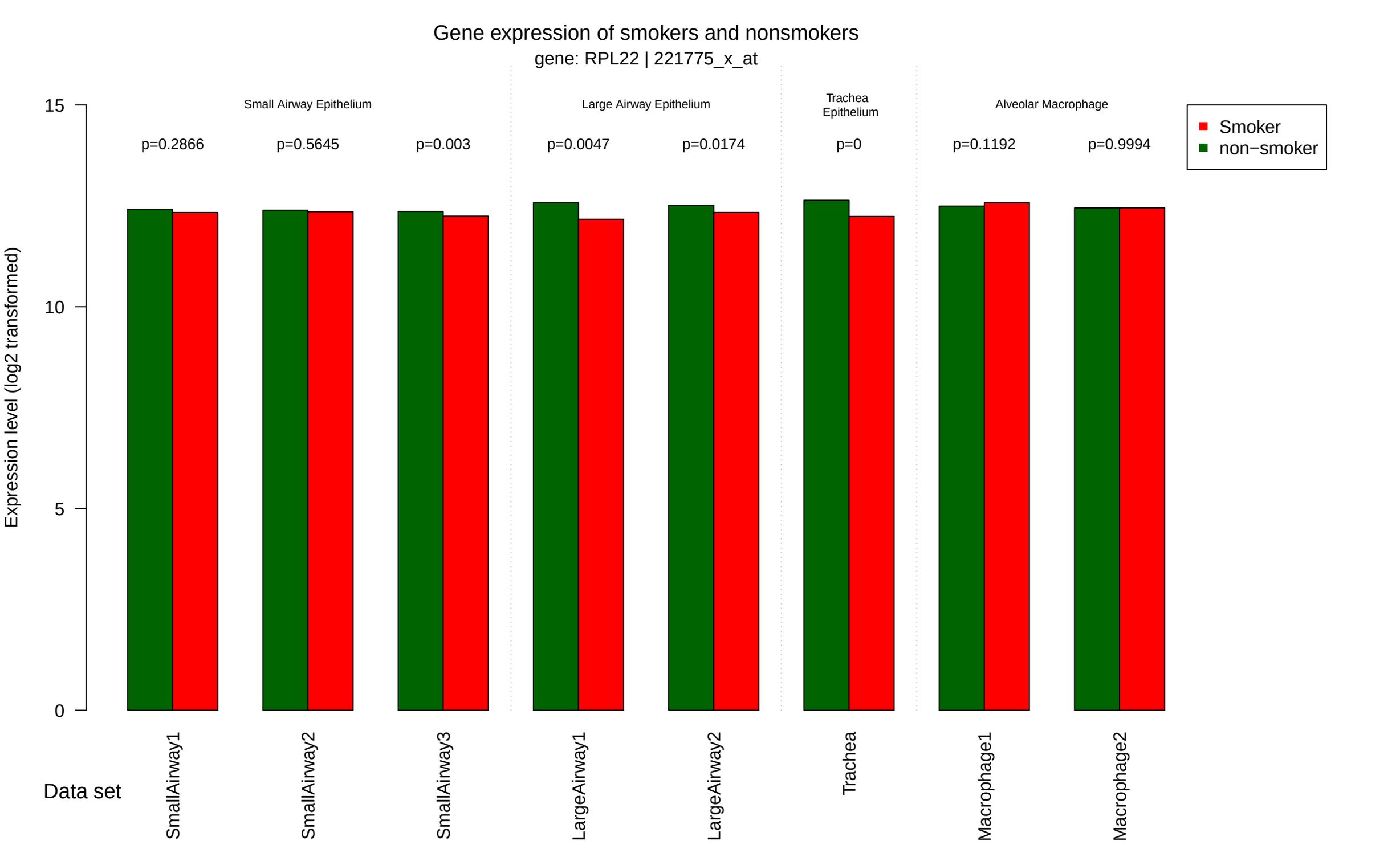
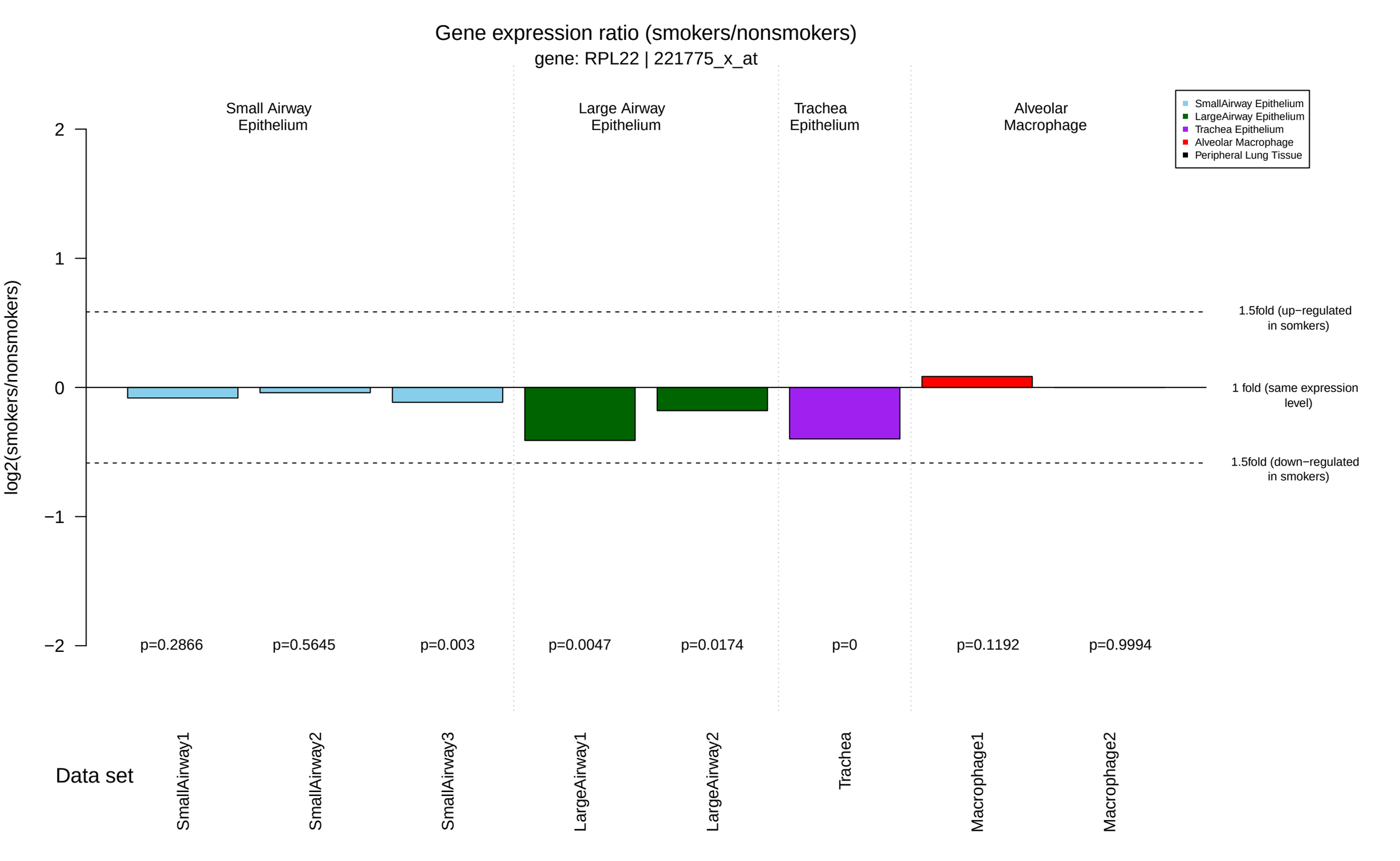
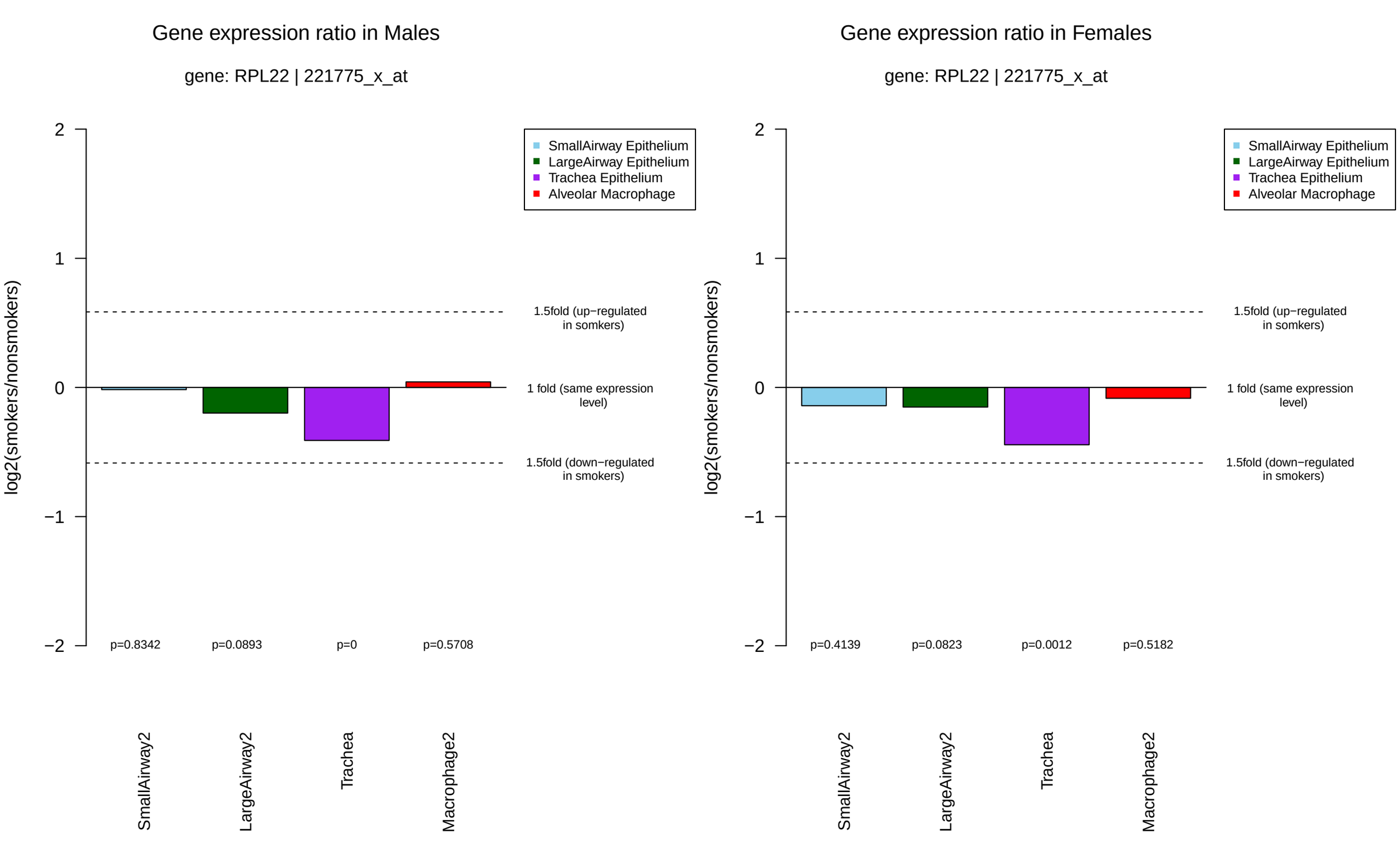
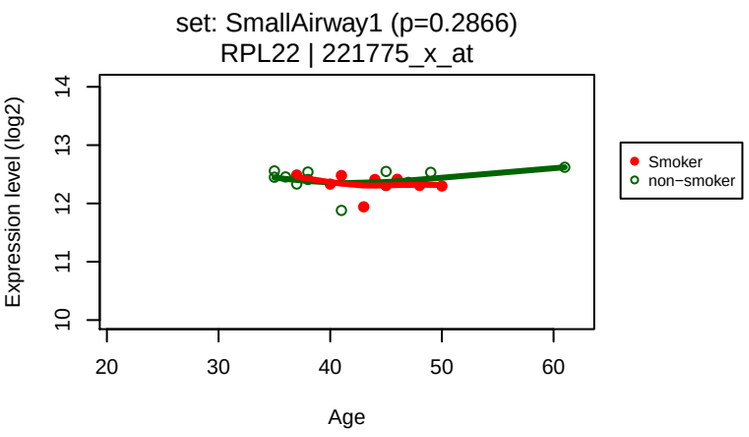
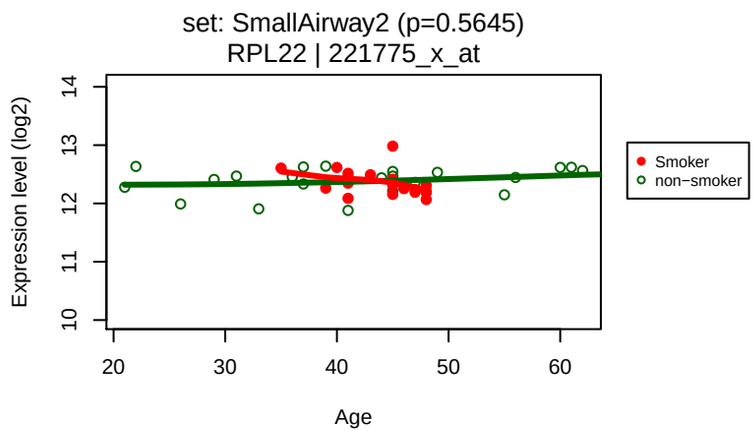
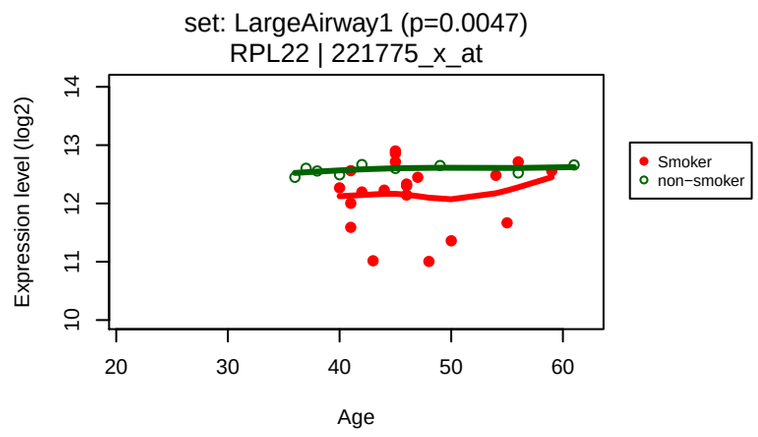

Gene expression of smokers and nonsmokers
gene: RPL22 | 221775_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2866
p=0.5645
p=0.003
p=0.0047
p=0.0174
p=0
p=0.1192
p=0.9994
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

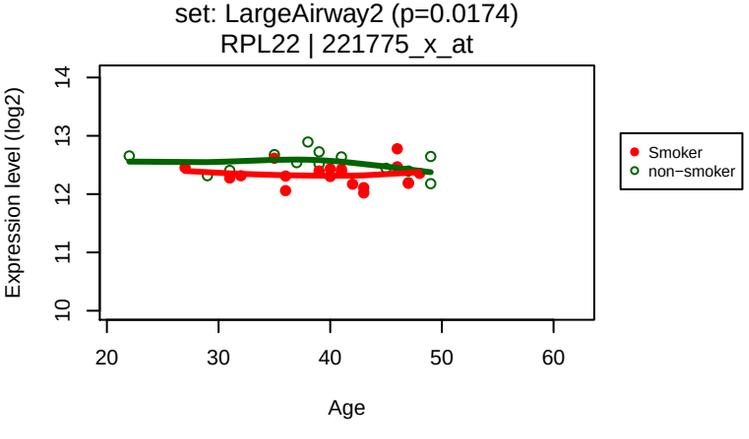

Gene expression ratio (smokers/nonsmokers)
gene: RPL22 | 221775_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2866
p=0.5645
p=0.003
p=0.0047
p=0.0174
p=0
p=0.1192
p=0.9994
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

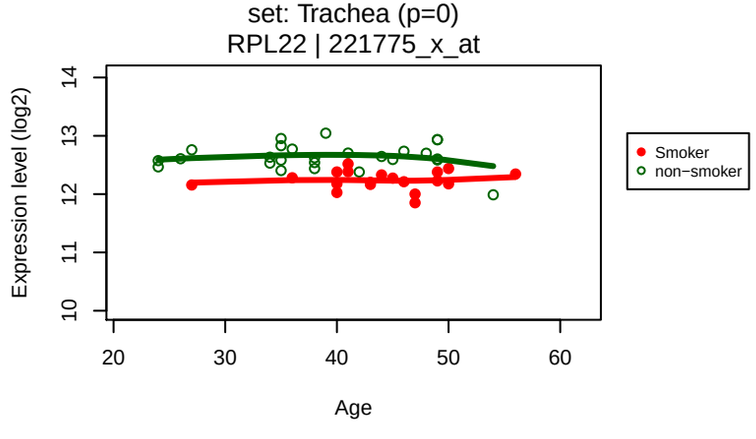

Gene expression ratio in Males
gene: RPL22 | 221775_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8342
p=0.0893
p=0
p=0.5708
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL22 | 221775_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4139
p=0.0823
p=0.0012
p=0.5182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

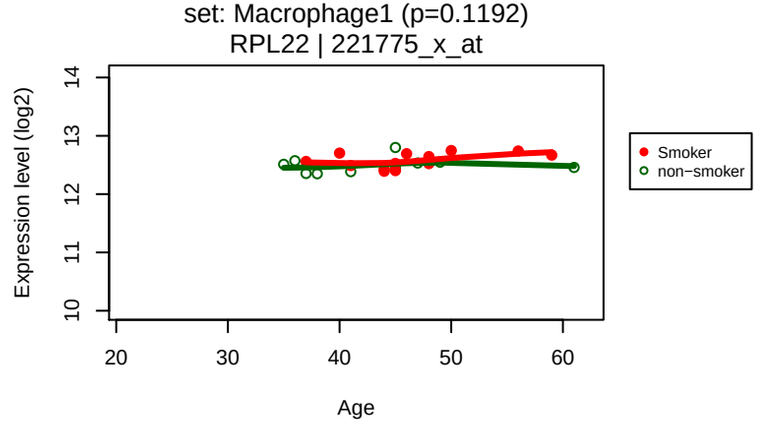

set: SmallAirway1 (p=0.2866)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker

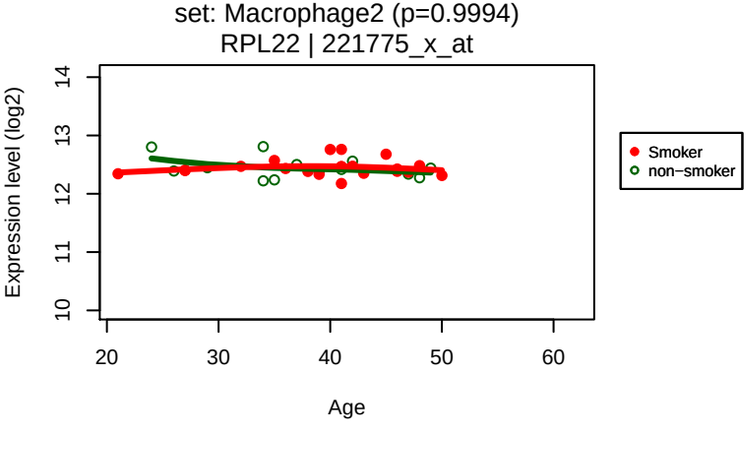

set: SmallAirway2 (p=0.5645)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker

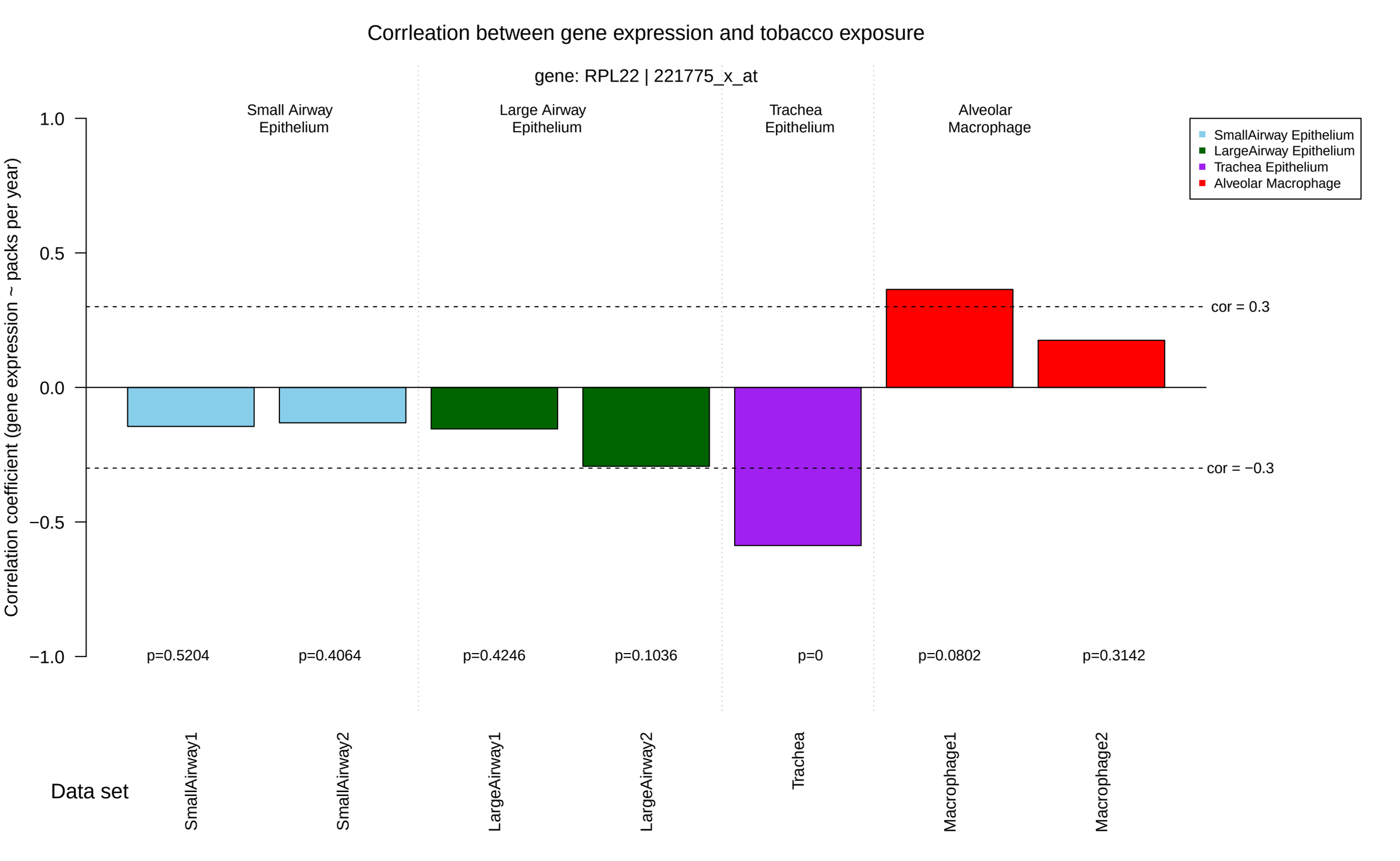

set: LargeAirway1 (p=0.0047)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0174)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1192)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9994)
RPL22 | 221775_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL22 | 221775_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5204
p=0.4064
p=0.4246
p=0.1036
p=0
p=0.0802
p=0.3142
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2